

main.py - Attendance Scenario Calculator

```
1 def get_int(msg, low=None, high=None):
2     while True:
3         val = input(msg)
4         try:
5             val = int(val)
6         except ValueError:
7             print("Please enter a valid number.")
8             continue
9         if low is not None and val < low:
10            print(f"Value should not be less than {low}.")
11            continue
12        if high is not None and val > high:
13            print(f"Value should not be more than {high}.")
14            continue
15        return val
16
17
18 def get_total_and_attended():
19     while True:
20         total = get_int("Total classes conducted: ", low=1)
21         attended = get_int("Classes attended: ", low=0)
22         if attended > total:
23             print("Attended classes can't be more than total classes.\n")
24             continue
25         return total, attended
26
27
28 def current_percentage(total, attended):
29     return (attended / total) * 100
30
31
32 def future_percentage(total, attended, extra_attend, extra_miss):
33     new_total = total + extra_attend + extra_miss
34     new_attended = attended + extra_attend
35     if new_total == 0:
36         return 0
37     return (new_attended / new_total) * 100
38
39
40 def classes_needed(total, attended, target):
41     curr = current_percentage(total, attended)
42     if curr >= target:
43         return 0
44
45     count = 0
46     t, a = total, attended
47     while (a / t) * 100 < target:
48         t += 1
49         a += 1
50         count += 1
51         if count > 100000:
52             break
53     return count
54
55
56 def option_current():
57     total, attended = get_total_and_attended()
58     percent = current_percentage(total, attended)
59     print(f"\nYour current attendance is {percent:.2f}%")
60
61
62 def option_target():
63     total, attended = get_total_and_attended()
64     target = get_int("Enter target percentage: ", low=0, high=100)
65
66     curr = current_percentage(total, attended)
67     print(f"\nCurrent attendance is {curr:.2f}%")
68
69     if curr >= target:
70         print("You already meet the target, no extra classes needed.")
71         return
72
73     n = classes_needed(total, attended, target)
74     print(f"You need to attend {n} more classes in a row to reach {target}%.")
75
76
77 def option_future():
78     total, attended = get_total_and_attended()
79     extra_attend = get_int("Upcoming classes you will attend: ", low=0)
80     extra_miss = get_int("Upcoming classes you will miss: ", low=0)
81
82     curr = current_percentage(total, attended)
83     fut = future_percentage(total, attended, extra_attend, extra_miss)
84
85     print(f"\nCurrent attendance: {curr:.2f}%")
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86     print(f"After {extra_attend} attended and {extra_miss} missed classes, "
87           f"attendance will be {fut:.2f}%")
88
89
90 def show_menu():
91     print("\n-----")
92     print("      ATTENDANCE SCENARIO CALCULATOR")
93     print("-----")
94     print("1. Calculate current attendance")
95     print("2. Find classes needed for target")
96     print("3. Predict future attendance")
97     print("4. Exit")
98
99
100 def main():
101     while True:
102         show_menu()
103         choice = input("Enter choice: ").strip()
104
105         if choice == "1":
106             option_current()
107         elif choice == "2":
108             option_target()
109         elif choice == "3":
110             option_future()
111         elif choice == "4":
112             print("\nThank you for using the calculator. Bye!")
113             break
114         else:
115             print("\nInvalid choice, try again.")
116
117
118 main()

```